



GoldStone Infratech's e-buses have hit Indian roads (Credit: <http://goldstoneebus.com>)



India needs more charging stations, and people be made more aware

ment's strategy to boost manufacturing in India. Accordingly, efforts and incentives are being given to promote the entire battery manufacturing ecosystem. International Solar Alliance (ISA) has reached out to South America's Lithium Triangle to procure metal used in lithium-ion batteries.

With all the premium being given to drive the segment, this is the best time to set up an EV manufacturing unit in India.

Business challenges

The major environmental barrier to this business is the toxic nature of lead. Also, there is competition from other types of batteries like fuel cell. Despite the various exemptions, India still remains a country that gives fewer incentives to its manufacturers as compared to other countries that are trying to develop an electric transport fleet.

Bruno expresses his concern, "Why are EV batteries taxed at 28 per cent GST? That is a big obstacle, both for businesses as well as customers. This needs to be fixed immediately."

Sood confirms, "Lithium-ion batteries are about 35 per cent costlier than lead-acid ones. This brings an impact on the price of the overall vehicle. If 28 per cent GST is added over this, it will be extremely difficult for businesses to make EVs at a reasonable price."

Samrath S. Kocchar, chief operating officer, Trontek Group, adds, "India has a lack of hardware production infrastructure and raw material for EV batteries. This impacts the domestic EV battery supply chain. This will not improve as long as products from China are made easily available. Unless there are more domestic players/manufacturers, it will not be possible to compete with the prices of Chinese batteries. That is a major concern for businesses who are planning to produce EV batteries in India."

He continues, "The level of financial investment required to start this business is quite high. However, if bigger players were to put investments in domestic battery manufacturing, small- and medium-scale players can follow. Unless manufacturing in India picks up, the sail will not smoothen."

Bruno adds, "Charging infrastructure in India is also inadequate. Unsatisfactory quality chargers impact longevity and performance of batteries, which, in turn, works against battery-driven vehicles adoption. Charging infrastructure needs huge investments. This will have to be driven majorly by the government, instead of private players."

What manufacturers need to do to succeed

Sood says, "A strong R&D department is important for any business to succeed in the EV battery market. Given the nascence of the field in India, there is a lot of scope in innovating on composition, materials and technology, and coming up with better cost-control processes."

Recently, Indian Space Research Organisation (ISRO) declared that it is in search, and discussions, for finding suitable industry partners to enable commercial EV battery manufacturing based on its space-age lithium-ion technology, which will substantially reduce battery costs.

To cope with the high upfront investment, industry experts suggest smaller companies to dive into the business after creating partnerships or clusters till the time a proper incentive scheme comes out.

Kocchar says, "Investments in EV are coming. Right now, India needs more incentives and subsidies from the government. For EV batteries, there have not been any incentives announced yet."

However, the situation may start to improve as a recent discussion between R.K. Singh, Union Minister of State (IC), Power and New & Renewable Energy, and various battery-industry players, explored an incentive for domestic battery production and recycling of lithium-ion batteries in India—another glaring challenge in the ecosystem today.

The discussion also revealed potential collaborations with countries like Bolivia to procure raw materials.

Kocchar says, "With all these collaborations taking centrestage, India's EV ecosystem and domestic manufacturing will exponentially improve. Next five to seven years are quite crucial."

—Paromik Chakraborty and Nidhi Arora, EFY

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